

RECAP

Introduction, Research Methods, Biological basis, Learning

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What is Psychology

- Psychology
 - Greek word: psyche = mind
 - Logos = knowledge or study
- Scientific study of behaviour and mental processes
 - Overt behaviours: directly observable actions and responses
 - Covert behaviours: internal
- Study of ABC
 - Affect: emotions; feelings; urges
 - Behaviour: (un)observable actions and responses taken by the individual
 - Cognitions: our inner thoughts; our rationales; our beliefs
- Not common sense:
 - Focuses on evidence
 - Systematic and intersubjective

Approaches to Psychology

- Father of Psychology: Wilhem Wundt (Structuralism)
- **Approaches:**
 - Biological: dopamine, reward pathway, prefrontal cortex
 - Behavioral: reinforcement, habit learning, cues
 - Humanistic: needs, meaning, self-growth
 - Cognitive: beliefs, attention, interpretation
 - Sociocultural: norms, peer influence, family context

Types of psychological professionals

🌟 Who is a psychiatrist? Who is a psychologist? 🌟

Type of professional	Psychiatrist	Psychologist	Psychiatric social worker
Educational degree	Doctor of Medicine (MD)	Doctorate degree	Master of Social Work (MSW)
Training	Medical doctor (MBBS)	Academic	Academic
Specialisation	▪ Diagnosis and treatment of psychological disorders ▪ May provide therapy and counselling	▪ Master's degree: counsellor, therapist, researcher ▪ Clinical psychologist: license by the Rehabilitation Council of India (RCI)	▪ Environmental conditions ▪ E.g., poverty, stress, drug misuse
Prescribe medications?	Yes	No	No
Workplace setting	Private practice or hospital settings	Multiple	Clinics, hospitals, social service organisations

Foundational Fields	Biological Psychology/Neuroscience	Examines the biological bases of behavior and mental processes
	Cognitive Psychology	Studies mental processes like memory, perception, and problem-solving
	Development Psychology	Examines psychological growth and changes throughout the lifespan
	Social Psychology	Explores how people influence and relate to each other
	Behavioural Psychology	Analyses observable behavior and its modification
	Personality Psychology	Investigates individual differences in personality traits
Applied Fields	Clinical	Focuses on diagnosing and treating mental health disorder
	Counselling	Helps individuals cope with personal and interpersonal challenges
	Health	Studies psychological factors affecting health and illness
	Industrial/Organisational	Focuses on human behavior in the workplace to improve productivity and well-being
	Educational	Focuses on how people learn and the effectiveness of educational interventions
	Forensic	Applies psychology to legal and criminal justice issues
	Sports	Optimises athletic performance and psychological well-being in sports
Interdisciplinary and emerging fields	Cultural Psychology	Cultural context shapes cognition, emotion, and behaviour
	Environmental	Studies the interaction between people and their physical environment
	Positive	Focus on well-being, strengths, resilience, flourishing
	Psycholinguistics	Focus on how we acquire and use language
	Neuropsychology	Brain and behaviour in a clinical setting

Research Methods

- Scientific Method
- Perceiving the question – Hypothesis – Hypothesis Testing – Conclusion – Report
- Types of research methods
 - Descriptive
 - Correlational
 - Experimental
- Descriptive types
 - Naturalistic observation: Observer bias
 - Laboratory observation
 - Clinical case study
 - Survey: Biases (Courtesy bias, Wording bias, Question order bias)

Cross sectional research

- Observational study that collects data from diverse population at a single point in time
- Helps understand prevalence and describe characteristics

Longitudinal research

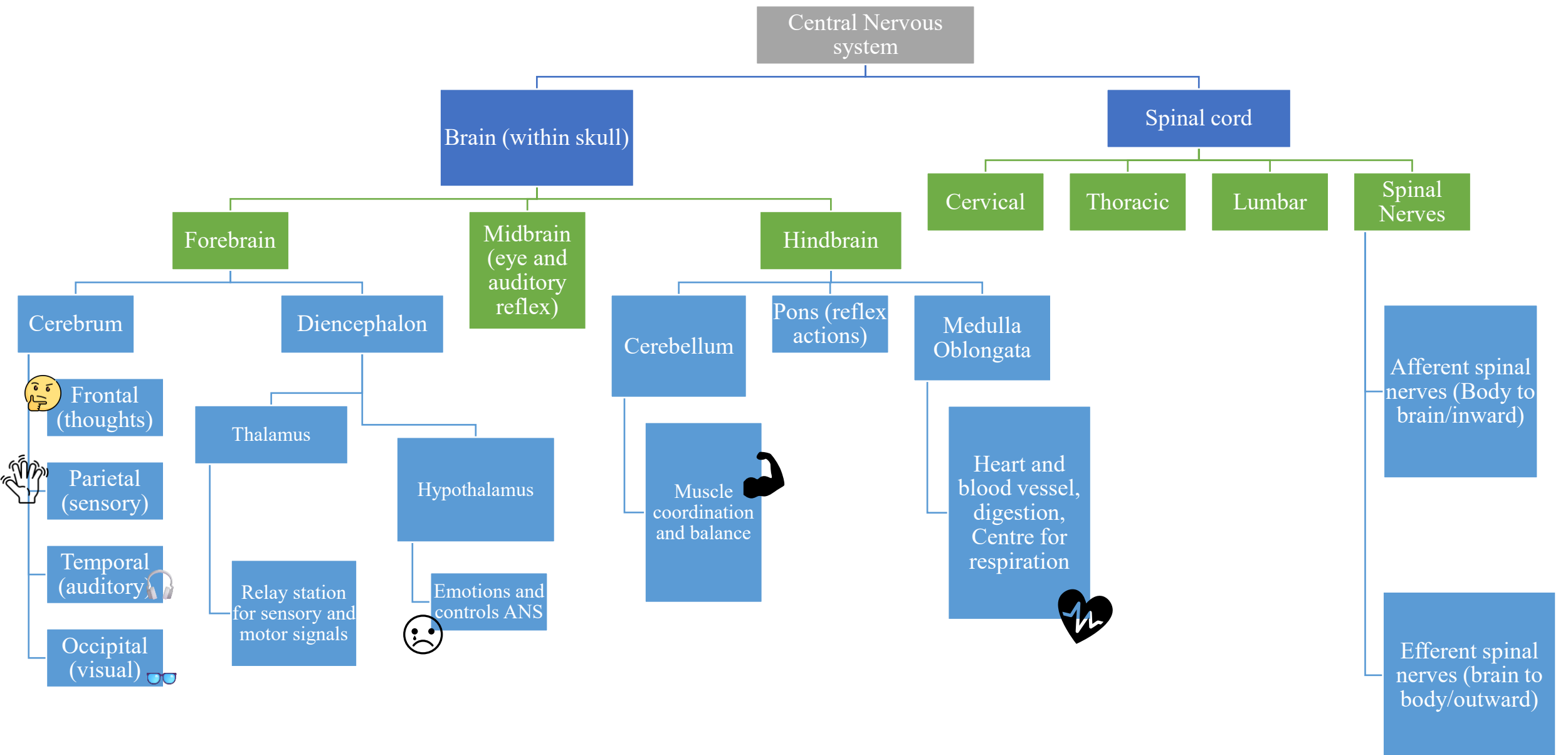
- A study method that involves collecting data from same individual or group over extended period of time from weeks to decades

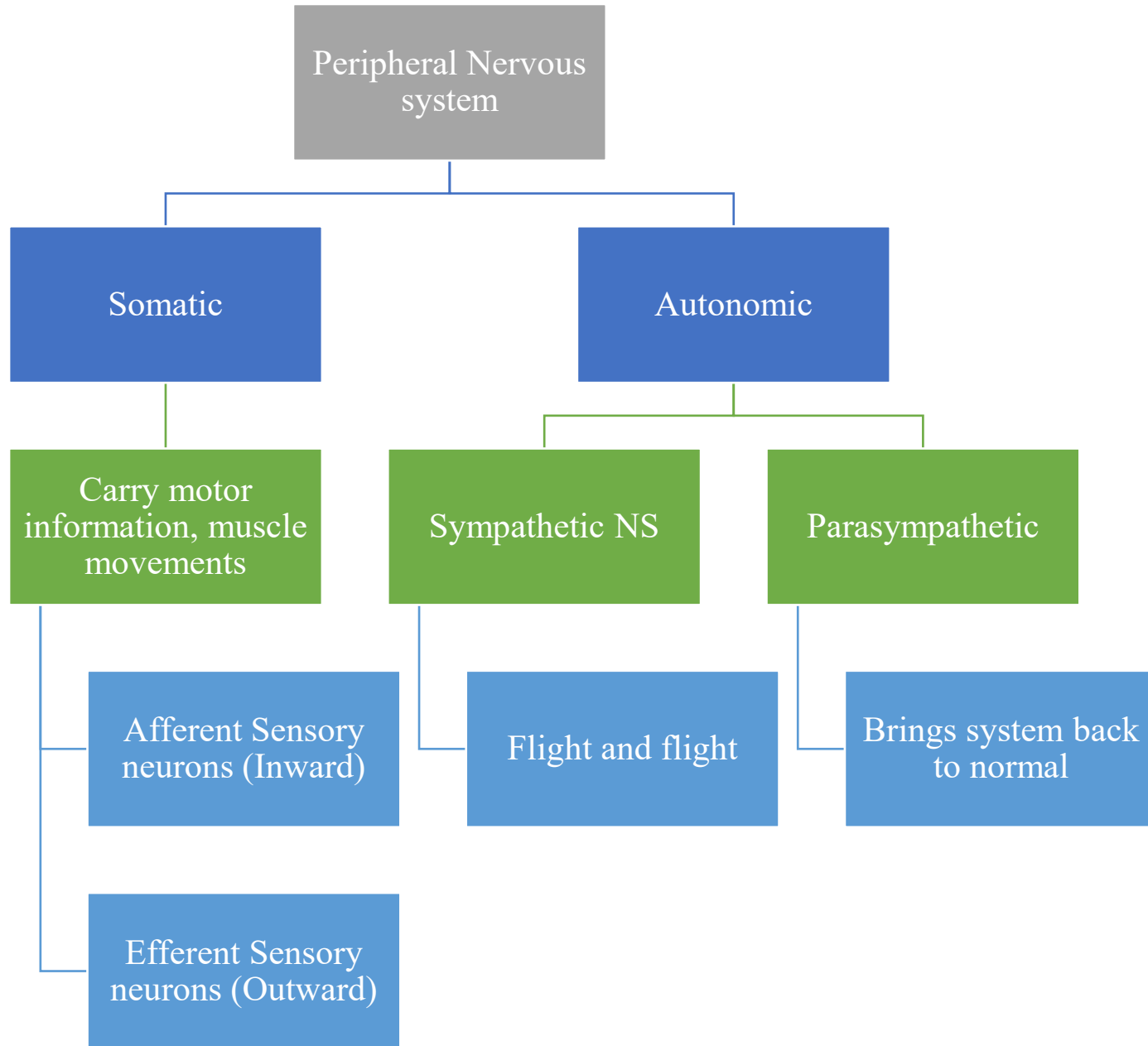
Correlational research

- Non experimental method that measures the statistical relationship between two or more variables
- It seen if change in one corresponds change in another without manipulation

Experimental Research

- Experiment: Direct manipulation of variables
- Variables
 - Independent
 - Dependent
 - Confounding
 - Extraneous
- Experimenter effect: unintentional influence
- Single anonymous study
- Double anonymous study





Forebrain

- Cortex
 - Frontal lobe: Broca Aphasia (difficulty in using expressive language)
 - Temporal lobe: Wernicke's aphasia (word salad/ difficult in word comprehension)
- Basal Ganglia (decision making and coordination)
 - Addiction
 - Alzheimer's
 - Parkinson's
 - Huntington's (Decay of brain nerve cells over time)
 - Tourette syndrome
- Limbic System
 - Thalamus (relay station for sensory except smell): Dyslexia
 - Hypothalamus (controls emotions): Hormones issues/ Depression
 - Cingulate cortex: Dyslexia (selective attention)
 - Hippocampus: Alzheimer (memory)
 - Amygdala: Klüver-Bucy syndrome

Midbrain and Hindbrain

- Midbrain: below Thalamus
 - Motor movements and reflexes
 - Damage: Parkinson's disease, issues with sleep
- Hindbrain
 - Medulla: Damage can be fatal
 - Pons: impaired coordination of movements
 - Cerebellum: Tremors unsteady walk (Alcoholic intoxication, Schizophrenia, Autism)

The Autonomic Nervous System

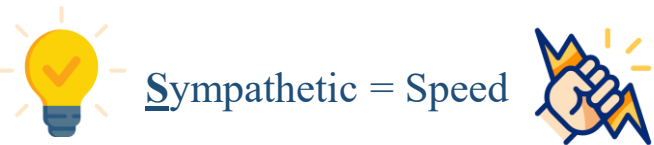
Sympathetic division

- Most active when you are angry, afraid, or aroused
- **Fight-or-flight response**
- Increases heart rate and breathing
- Stops digestion

Sympathetic division (arousing)		Parasympathetic division (calming)
Pupils dilate	EYES	Pupils contract
Decreases	SALIVATION	Increases
Perspires	SKIN	Dries
Increases	RESPIRATION	Decreases
Accelerates	HEART	Slows
Inhibits	DIGESTION	Activates
Secrete stress hormones	ADRENAL GLANDS	Decrease secretion of stress hormones

Parasympathetic division

- **Calms body**
- Produces effects opposite to those of the sympathetic division
- Reduces heart rate and breathing
- Restores digestion



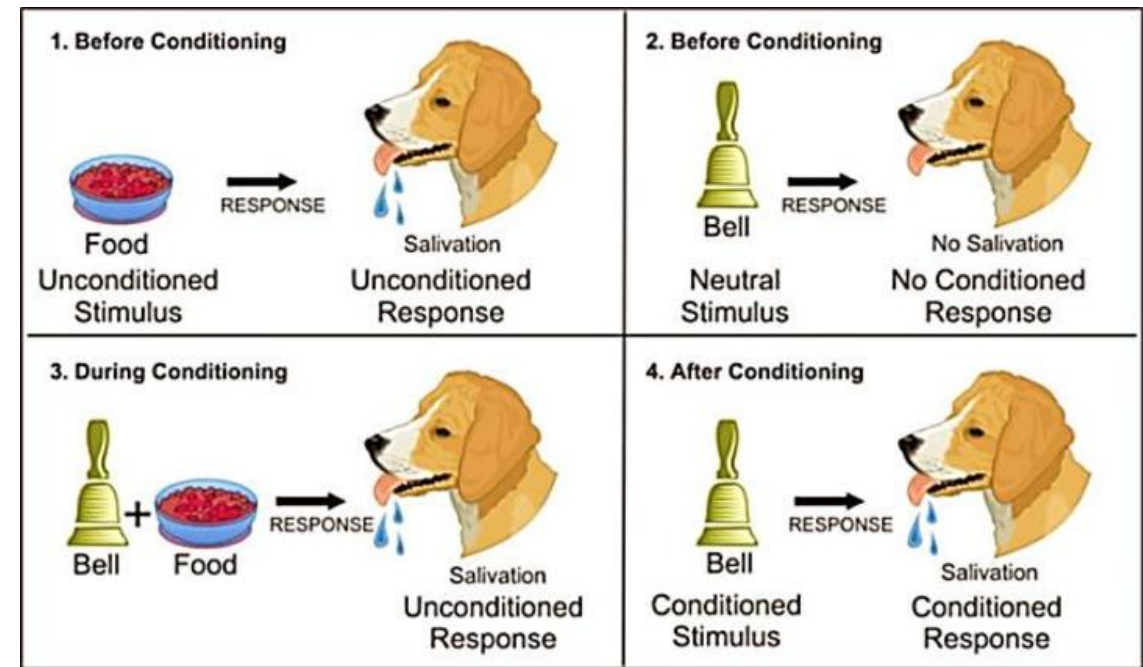
Sympathetic = Speed

Neuroplasticity

- Brains ability to change the structure and function in response to experience, learning, environment, and injury
- Types
 - Experience dependent plasticity
 - Developmental plasticity
 - Injury-induced plasticity

Learning

- Classical conditioning
 - Involuntary response
 - Stimulus generalization (similar stimuli)
 - Stimulus discrimination (respond to specific stimuli)
- Operant conditioning
 - Voluntary response
- Observational learning
 - Learning by observation
 - Attention, Memory, Imitation, Desire



	Reinforcement	Punishment
Positive (Adding)	Positive Reinforcement Something valuable or desirable is added	Punishment by Application Something unpleasant is added
Negative (Removing/avoiding)	Negative reinforcement Something unpleasant is removed	Punishment by removal Something desirable is removed